

Market Requirements Document (MRD)

Perfect Purifiers LLC | Product Development Planning | Written by Brian McCarthy

Market Requirements Document (MRD)

Section	Contents
1. Market Segment	Targeted market segments: (1) households with two or more adults, especially buyers over age 35; (2) families and health-conscious consumers in homes, apartments, and communities affected by allergens, smoke, odor, or hazardous summer air conditions; (3) small offices and wellness-oriented light commercial spaces that need broader coverage and safer operation. Specific buyers include homeowners, renters, parents, caregivers, office managers, and retail/online consumers. Specific users include adults, children, elderly occupants, remote workers, employees, guests, and customers sharing indoor spaces.
2. Market Problem	Customers want safer and more complete air purification than many current HEPA products provide. Key problems include inability to address VOCs, viruses, and odors; potential ozone emissions; noisy operation; short filter life and expensive annual replacements; maintenance burden that can lead to mold and bacteria buildup; limited room coverage; and price sensitivity for larger coverage solutions. These problems reduce satisfaction, increase returns, and create a gap in the current HEPA market.
3. Market Requirements	Customers require a purifier that reduces as many contaminants as possible, operates safely with zero or minimal ozone, reduces odors, runs quietly, covers a wider area, requires less frequent and easier maintenance, uses a lower-cost filter with longer useful life, and is available at multiple price points. They also require clear product positioning so buyers can understand Basic, Better, and Ultimate differences before purchase.
4. Prioritization	Requirement priority is based on customer importance, safety, competitive differentiation, feasibility, and launch timing. Must Have: ozone elimination or significant reduction; 50% noise reduction; improved motor coverage; clear model tiers. Should Have: longer-life 1.5-year filter; lower-cost easy-replacement filter; odor/air freshener and cleaning option. Could Have: additional commercial-grade features after launch. Won't Have for initial launch: heavy-duty wet/dry scrubber expansion and highly customized industrial configurations.

Prioritized Requirements Detail

Rank	Requirement	Priority	Reason
1	Eliminate or significantly reduce ozone generation	Must Have	Directly addresses safety, health, environmental concerns, and customer refusal to purchase ozone-emitting products.
2	Reduce purifier noise by 50%	Must Have	Improves daily usability for bedrooms, living rooms, offices, and remote work settings.
3	Increase filtration coverage by 50% through improved motor strength	Must Have	Solves the small-area coverage complaint and supports stronger value at each model tier.
4	Offer three product tiers: Basic, Better, Ultimate	Must Have	Creates price flexibility and supports multiple buyer segments with clear value propositions.
5	Introduce lower-cost, easy-replace filter with 1.5-year life	Should Have	Reduces total cost of ownership and maintenance friction.
6	Add air freshener and cleaning option	Should Have	Addresses odor and perceived freshness, differentiating the Ultimate model.
7	Prepare wholesale, retail, and online channel launch support	Should Have	Uses existing distribution logistics and supports the six-month market window.